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In [14]: import pandas as pd
import numpy as np

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import matplotlib.pyplot as plt
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```
In [15]: df = pd.read_csv(r"D:\TE\A37\dsBDA\mini_Project\movie_dataset.csv")
```

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In [16]: df
```

Out[16]:

	index	budget	genres	homepage	
0	0	237000000	Action Adventure Fantasy Science Fiction	http://www.avatarmovie.com/	199
1	1	300000000	Adventure Fantasy Action	http://disney.go.com/disneypictures/pirates/	2
2	2	245000000	Action Adventure Crime	http://www.sonypictures.com/movies/spectre/	2066
3	3	250000000	Action Crime Drama Thriller	http://www.thedarkknightises.com/	490
4	4	260000000	Action Adventure Science Fiction	http://movies.disney.com/john-carter	495
...	...	...	...	...	...
4798	4798	220000	Action Crime Thriller		NaN 93
4799	4799	9000	Comedy Romance		NaN 727
4800	4800	0	Comedy Drama Romance TV Movie	http://www.hallmarkchannel.com/signedsealeddel...	2316
4801	4801	0	NaN	http://shanghaicalling.com/	1261


```
In [19]: def combine_features(row):  
         return row['keywords'] + " " + row['cast'] + " " + row['genres'] + " " + row['d  
  
df["combined_features"] = df.apply(combine_features, axis=1)
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In [20]: cv = CountVectorizer()  
count_matrix = cv.fit_transform(df["combined_features"])
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In [21]: cosine_sim = cosine_similarity(count_matrix)
```

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In [22]: def get_title_from_index(index):  
         return df.iloc[index]["title"]  
  
def get_index_from_title(title):  
    return df[df.title == title].index[0]
```

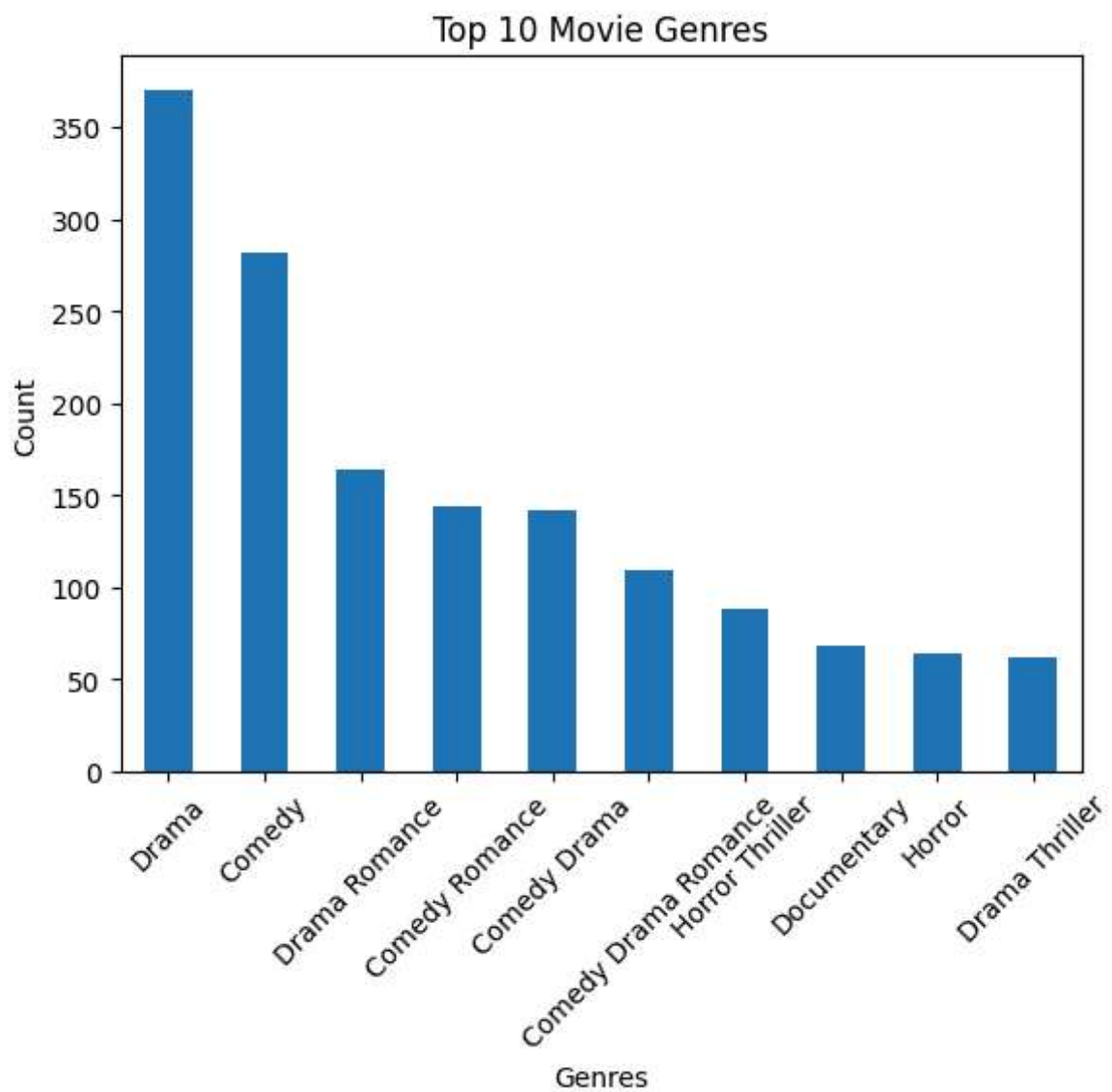
```
In [23]: def recommend(movie_name):  
         movie_index = get_index_from_title(movie_name)  
         similar_movies = list(enumerate(cosine_sim[movie_index]))  
  
         sorted_movies = sorted(similar_movies, key=lambda x: x[1], reverse=True)  
  
         print("Top 5 movies similar to", movie_name, ":\n")  
  
         for movie in sorted_movies[1:6]:  
             print(get_title_from_index(movie[0]))
```

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In [24]: recommend("Spectre")
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Top 5 movies similar to Spectre :

Skyfall
Quantum of Solace
The Girl with the Dragon Tattoo
The Hunger Games: Catching Fire
Johnny English Reborn

```
In [26]: df['genres'].value_counts().head(10).plot(kind='bar')  
plt.title("Top 10 Movie Genres")  
plt.xlabel("Genres")  
plt.ylabel("Count")  
plt.xticks(rotation=45)  
plt.show()
```



In []: